

Algebra 1: Unit 11, Lesson 6

1. Pablo opened a savings account for college with \$1,200. He let it grow for 10 years with an annual earnings rate of 3.75%.

What is the exponential equation that represents this situation?

2. A new car costs \$37,500. The value depreciates each year by 12%. What is the value of b in the form $y = a(b)^x$ that represents this situation?

3. Anne found a spot of mold on her bread. She decided to see how quickly it spread over the course of three days. When she found it, the spot of mold was 2 millimeters (mm) wide. The second day, it was 6 mm wide. The third day, it was 18 mm wide. This can be modeled by the function $y = a(3)^x$.

What is the a value for this function?